

Final Technical Review for HaskLedger

Review Details

Reviewer name: Koz Ross

Role: Consultant, head developer for Covenant

Areas reviewed: HaskLedger's implementation in Haskell, documentation

Approximate time spent: 2 hours

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Overall technical assessment

The HaskLedger project is a capable and interesting framework for Cardano scripts. It gives significant performance benefits over other frameworks for Plutus scripting, and makes use of a novel intermediate and back-end in Covenant. While it is still new and could use improvements from the UX side, it is usable and useful today.

Key strengths

Performance is shown to be very strong. The eDSL provided is intuitive and easy to follow. The separation of front-end and back-end allows improvements independently of each other, allowing for easier maintenance and improvements, such as performance upgrades and support for new versions of Plutus.

Key concerns and limitations

Currently, the user-facing documentation is sparse. While Haddocks for HaskLedger itself are provided, and examples, along with some explanations, are given, users of HaskLedger would need to spend considerable time reading code to see what they would get. This makes HaskLedger less convenient than, for example, Aiken would be currently. Additional clarity about compilation strategy and outputs would also be useful, especially in light of Covenant's somewhat novel approach.

Recommended improvements

HaskLedger could benefit from additional documentation, particularly explaining its approach to specific aspects of code generation (onchain representations, how things compile, etc), as well as some helper tools similar to those provided by Plutarch and Aiken, such as for testing. Essentially, the HaskLedger framework should aim to have documentation and tooling to the level of Aiken.

Specific feedback items requiring response and action

No specific actions required at this time, though the recommended improvements given above are a good long-term target.

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